

National University of Computer & Emerging Sciences

Quiz # 2(a)

Name & Roll Number: _____

Section: _____

Date: _____

Maximum Marks 10

1. A professor records the execution times (in milliseconds) of a **merge sort algorithm** on different input sizes. The recorded times are:

5, 10, 15, 22, 28, 33, 38, 41, 47, 52

- (a) Find the **mean**, **median** and **mode** execution time.
- (b) If a new execution time of **55 ms** is added, how does it affect the mean? (Increase, decrease, or stay the same?)
- (c) Compute the **range** and **standard deviation** of execution times.
- (d) Find the **1st quartile (Q1)** and **3rd quartile (Q3)** of the execution times.
- (e) What is the **percentile rank** of **33 ms** in the dataset?

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Quiz # 2(b)

Name & Roll Number: _____

Section: _____

Date: _____

Maximum Marks 10

1. A CS student is analyzing the **response times (in milliseconds)** of a web server under different loads. The recorded times are:

120, 135, 150, 165, 180, 195, 210, 225, 240, 255

- (a) Find the **mean**, **median** and **mode** response time.
- (b) If a new response time of **270 ms** is added, how does it affect the mean? (Increase, decrease, or stay the same?)
- (c) Compute the **range** and **standard deviation** of response times.
- (d) Find the **1st quartile (Q1)** and **3rd quartile (Q3)** of the response times.
- (e) What is the **percentile rank** of **195 ms** in the dataset?

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Quiz # 2(c)

Name & Roll Number: _____

Section: _____

Date: _____

Maximum Marks 10

1. A machine learning engineer records the **processing times (in seconds)** for different deep learning models. The recorded times are:

2.1, 2.5, 3.0, 3.7, 4.4, 5.2, 6.1, 7.3, 8.6, 10.0

- (a) Find the **mean**, **median** and **mode** processing time.
- (b) If a new processing time of **11.5 seconds** is added, how does it affect the mean? (Increase, decrease, or stay the same?)
- (c) Compute the **range** and **standard deviation** of processing times.
- (d) Find the **1st quartile (Q1)** and **3rd quartile (Q3)** of the processing times.
- (e) What is the **percentile rank** of **5.2 seconds** in the dataset?

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Quiz # 2(d)

Name & Roll Number: _____

Section: _____

Date: _____

Maximum Marks 10

1. A network engineer is analyzing **packet transmission times (in milliseconds)** across different network nodes. The recorded times are:

30, 35, 40, 42, 48, 53, 57, 60, 65, 70

- (a) Find the **mean**, **median** and **mode** transmission time.
- (b) If a new transmission time of **75 ms** is added, how does it affect the mean? (Increase, decrease, or stay the same?)
- (c) Compute the **range** and **standard deviation** of transmission times.
- (d) Find the **1st quartile (Q1)** and **3rd quartile (Q3)** of the transmission times.
- (e) What is the **percentile rank** of **53 ms** in the dataset?